

# Akhil Vanukuri

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## Research Interests

Cryptography, Security.

## Relevant Experience

Worked on theory of cryptography and also implemented proof of concept of a cryptographic protocol as part of two different projects (details below).

## Education

|                                              |                                                                                                                                          |                        |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>PhD in Computer Science</b>               | University of Wisconsin, Madison<br>CGPA: 3.893 / 4.0                                                                                    | Aug, 2024 - present    |
| <b>MS (By Research) in C.S.E<sup>1</sup></b> | Indian Institute of Technology, Madras<br>Advisor: Shweta Agrawal<br>Thesis: Cryptographic applications of $k$ -SUM<br>CGPA: 8.80 / 10.0 | Jan, 2023 - July, 2024 |
| <b>MSc. in Computer Science</b>              | Chennai Mathematical Institute<br>Thesis advisor: Shweta Agrawal<br>Thesis: Fully Homomorphic Signatures<br>CGPA: 8.81 / 10.0            | Dec, 2020 - Jul, 2022  |
| <b>B.Tech. in C.S.E</b>                      | Manipal Institute of Technology, Manipal<br>CGPA: 8.22 / 10.0                                                                            | Jul, 2016 - Aug, 2020  |

## Conference Publications

**$k$ -SUM in the Sparse Regime**  
Shweta Agrawal, Sagnik Saha, Nikolaj I. Schwartzbach, [Akhil Vanukuri](#), Prashant Nalini Vasudevan  
Accepted at CRYPTO'24: The 44<sup>th</sup> Annual International Cryptology Conference

## Experience

|                            |                                                                                                    |                     |
|----------------------------|----------------------------------------------------------------------------------------------------|---------------------|
| • <b>Project Associate</b> | Indian Institute of Technology, Madras                                                             | May 2022 - May 2024 |
| • <b>Research Intern</b>   | National University of Singapore, Singapore<br>Internship Supervisor: Prashant Nalini Vasudevan    | Jun 2023 - Aug 2023 |
| • <b>Student Trainee</b>   | Samsung Research and Development Institute, Bangalore                                              | Jan 2020 - Jun 2020 |
| • <b>Research Intern</b>   | R. C. Bose Centre for Cryptology, and Security, ISI Kolkata<br>Internship Supervisor: Sushmita Ruj | May 2019 - Jul 2019 |

## Professional Activities

Sub-reviewer for Crypto 2023, PKC 2024, Crypto 2025.

## Research Projects

- **Study of  $k$ -SUM problem in the sparse regime** IIT Madras and NUS, Singapore  
Studied properties of the  $k$ -SUM problem in the sparse regime and constructed cryptographic schemes using it

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<sup>1</sup>Computer Science and Engineering

- **Cloud Data Auditing Using Proofs of Retrieval and Replication** ISI, Kolkata  
Implemented proof of concept (using gmp, pbc library in C) and made contributions w.r.t mathematical aspects of a public verifiable privacy-preserving proof of replication scheme  
(Co-author of a manuscript with Nishant Dhanaji Nikam and Sushmita Ruj).
- **Dehazing** Samsung R & D, Bangalore  
Designed and implemented a novel CNN-based solution for dehazing of a hazy image (using PyTorch).

## Scholarships

- 2020-2022    Cognizant Foundation Scholarship

## Talks

- **$k$ -SUM in the sparse regime** Bangalore Crypto Day, 2024

## Relevant Courses<sup>2</sup>

Structure vs Hardness in Cryptography  
Grade : 10/10

Computability and Complexity  
Grade : 9/10

Foundations of Cryptography  
Grade : 9/10

Quantum Algorithms and Cryptography<sup>3</sup>  
Grade : 10/10

Topics in Cryptography  
Grade : 9/10

Algebra II (Abstract Algebra)  
Grade : 8/10

Modern Complexity Theory  
Grade : 8/10

User-Centric Security & Privacy  
Grade : AB

Introduction to Information Security  
Grade : A

Computer Networks  
Grade : 9/10

Linear Algebra (Accelerated Honors)  
(Audited)

Structural Graph Theory  
Grade : 8/10

## Teaching Assistance

- **CS1100: Introduction to Programming**  
Lead Teaching Assistant for both theory and lab component of the course at IIT, Madras.  
*Awarded Star TA award for the work done.*
- **CS577: Introduction to Algorithms**  
Teaching Assistant at UW-Madison
- **CS435: Introduction to Cryptography**  
Teaching Assistant at UW-Madison
- **CS240: Discrete Mathematics**  
Teaching Assistant at UW-Madison

## Other Projects <sup>4</sup>

- **Parallelization of AES encryption**  
Implemented a parallelized version of AES using CUDA, OpenGL.

<sup>2</sup>Click on the subjects to see the topics covered

<sup>3</sup>under the name "Quantum Algorithms and Complexity" in the transcript

<sup>4</sup>These projects are part of undergrad course-requirement and have been done in collaboration with other classmates.

- **Placement Portal Website**

Developed a placement management website using C# and MySQL.

- **Parser for Lua**

Developed a parser for the programming language Lua using C, Flex, Bison.

- **Movie Database App**

Developed an application that allows information retrieval from a database storing film-related information using Java, SQL.

## Miscellaneous

- Programming Languages: C, JAVA, Python, Haskell, SQL
- Other relevant technical skills: CUDA, OpenGL, MPI, LaTeX, HTML, Git.
- Libraries: GMP (GNU Multiple Precision Arithmetic), PBC (Pairing Based Crypto), PyTorch